

Dysostosis Multiplex, Ain-Naz Type (DMAN): A Comprehensive Disease Report

Disease: Dysostosis Multiplex, Ain-Naz Type (DMAN) **MONDO ID:** MONDO:0859156 | **OMIM:** #619345 | **MedGen:** C5444223 **Causal gene:** *LYSET* (formerly *TMEM251*; alias *GCAF*) — HGNC:20218, 14q32.12 **Category:** Mendelian, autosomal recessive lysosomal storage disorder (mucopolidosis-like)

Summary

Dysostosis Multiplex, Ain-Naz Type (DMAN) is an **ultra-rare, autosomal-recessive skeletal dysplasia** caused by **biallelic loss-of-function variants in *LYSET*** (formerly *TMEM251*, alias *GCAF*), a small Golgi-resident two-transmembrane protein. The disease was first defined in 2020–2021 by whole-exome sequencing of two unrelated consanguineous families, and the clinical picture is that of a **mucopolysaccharidosis (MPS)-like dysostosis multiplex** with extreme short stature (−8 to −11 SD), progressively coarse facial features, protruding abdomen, progressive skeletal deformity, loss of mobility, and premature death (two oldest reported patients died in their twenties; one child died at age 5).

Mechanistically, DMAN is a **mannose-6-phosphate (M6P) pathway disorder** that is biochemically and clinically convergent with **mucopolidosis II/III (I-cell disease)**. *LYSET* is required to stabilize and retain **GlcNAc-1-phosphotransferase (GNPT)** in the *cis*-Golgi. When *LYSET* is lost, GNPT — which has a hydrophilic transmembrane domain — becomes unstable and is mislocalized from the Golgi to the lysosome. This abolishes **M6P tagging** of the ~60 soluble lysosomal hydrolases, so these enzymes are **hyper-secreted into plasma rather than trafficked to lysosomes**, producing an intracellular lysosomal-storage state. The result is the classic "I-cell" biochemical signature: **elevated lysosomal enzymes in plasma, reduced enzymes in fibroblasts**, and abnormal urinary glycosaminoglycans and sialic acid. For this reason, functional groups have proposed the alternative disease names "**LYSET-related mucopolidosis**" and "**Mucopolidosis Type V**," and classify the disorder among the "**Golgipathies**."

There is **no disease-specific cure**. Management is supportive, and candidate disease-modifying strategies are extrapolated from mucopolysaccharidosis II research: hematopoietic stem cell transplantation (partial biochemical benefit but skeletal/airway disease progresses), bisphosphonates (zoledronic acid partially restored hydrolase activity and reduced storage in an MLII mouse model), enzyme cross-correction using M6P-tagged supernatant, and variant-specific ASO exon-skipping. A **zebrafish *tmem251/lyset* knockout** phenocopies mucopolysaccharidosis II (heart edema, skeletal dysplasia) and serves as the principal validated animal model.

Section 1 — Disease Information

Overview. DMAN is a Mendelian, autosomal-recessive skeletal dysplasia resembling a mucopolysaccharidosis. Affected individuals present in infancy/early childhood with severe short stature, progressively coarse ("gargoyle-like") facies, protruding abdomen, and a radiographic pattern of **dysostosis multiplex**, with progressive skeletal deformity and loss of mobility ([PMID: 33252156](#)).

Key identifiers.

Resource	Identifier
MONDO	MONDO:0859156
OMIM (phenotype)	#619345 (Dysostosis Multiplex, Ain-Naz Type)
OMIM (gene)	619332 (LYSET/TMEM251*)
MedGen	C5444223
Gene / HGNC	LYSET (TMEM251), HGNC:20218
Ensembl	ENSG00000153485
UniProt	Q8N6I4
ICD-10 / ICD-11	No specific code; maps to skeletal dysplasia / lysosomal storage disorder categories
MeSH	No dedicated term; related to "Mucopolidoses," "Osteochondrodysplasias"

Synonyms / alternative names. - Dysostosis Multiplex, Ain-Naz Type (DMAN) — OMIM designation - **LYSET-related mucolipidosis** (proposed; [PMID: 40171858](#)) - **Mucolipidosis Type V** (proposed from zebrafish/functional work; [PMID: 36096887](#)) - Severe skeletal dysplasia with extreme short stature, TMEM251-related

Data source type. Information is derived from **individual-patient reports** (aggregated across ~4 families / 6+ patients described worldwide), combined with **disease-level/mechanistic resources** (OMIM, functional cell-biology and CRISPR-screen studies). This is an ultra-rare disease with no registry-scale epidemiology.

Section 2 — Etiology

Primary cause (genetic). DMAN is caused by **biallelic (homozygous or compound-heterozygous) loss-of-function variants in *LYSET*/TMEM251**. The founding report identified two homozygous variants in two unrelated consanguineous families: **c.133C>T; p.(Arg45Trp)** (missense) and **c.215dupA; p.(Tyr72Ter)** (frameshift/nonsense) ([PMID: 33252156](#)):

"Whole-exome sequencing identified two homozygous variants c.133C>T; p.(Arg45Trp) and c.215dupA; p.(Tyr72Ter), respectively, in the two families, affecting an evolutionary conserved gene TMEM251 (NM_001098621.1)."

Genetic risk factors. The only established risk factor is **inheritance of two pathogenic *LYSET* alleles**. **Consanguinity** is a major contributing circumstance — the initial families were consanguineous, and subsequent patients (Iranian brothers) were also from consanguineous unions, consistent with a rare recessive founder/homozygosity pattern.

Environmental risk factors. None identified. This is a monogenic Mendelian disorder with no known environmental, toxic, infectious, occupational, lifestyle, dietary, age, or sex-based risk factors contributing to causation.

Protective factors. No genetic or environmental protective factors have been described. In principle, any residual GNPT activity (e.g., hypomorphic alleles) would be expected to attenuate severity, analogous to mucolipidosis III versus II, but this has not been formally demonstrated for *LYSET*.

Gene–environment interactions. None reported; DMAN penetrance appears driven entirely by genotype.

Section 3 — Phenotypes

The core phenotype is an **MPS-like dysostosis multiplex** with early onset and progressive course ([PMID: 33252156](#)):

"affected individuals had a dysostosis multiplex-like skeletal dysplasia and severe short stature (<-8.5 SD). They manifested increasingly coarse facial features, protruding abdomens, and progressive skeletal changes, reminiscent of mucopolysaccharidosis. The patients gradually lost mobility and the two oldest affected individuals died in their twenties."

Additional documented features include craniosynostosis, kyphoscoliosis/scoliosis, hemivertebrae, hip subluxation/dislocation, dental abnormalities, and a ventricular septal defect (VSD) in one patient. The first Western (Brazilian) patient additionally had **valvular disease** (mitral/aortic valve thickening, tricuspid regurgitation), **hand contractures**, and **coarse facies from 5 months of age**; she died in 2018 from respiratory causes ([PMID: 41858182](#)).

Phenotype table with suggested HPO terms

Phenotype	Type	Onset	Severity / Course	Frequency	Suggested HPO
Dysostosis multiplex	Physical / radiographic	Infancy/ early childhood	Severe, progressive	Core (all)	HP:0000943
Severe short stature (−8 to −11 SD)	Physical	Infancy	Severe, progressive	Core (all)	HP:0004322
Progressive coarse facial features	Clinical sign	Infancy (from ~5 mo)	Severe, progressive	Core (all)	HP:0000280
Protruding abdomen	Clinical sign	Childhood	Moderate	Frequent	HP:0001538
Kyphoscoliosis / scoliosis	Physical	Childhood	Progressive	Frequent	HP:0002751 / HP:0002650
Hemivertebrae	Skeletal	Congenital	—	Reported	HP:0002937
Craniosynostosis	Skeletal	Congenital/ infancy	—	Reported	HP:0001363
Hip subluxation/ dislocation	Skeletal	Childhood	Progressive	Reported	HP:0002827
Loss of ambulation / joint contractures	Physical	Progressive	Severe	Frequent	HP:0002505 / HP:0002803
Dental abnormalities	Physical	Childhood	—	Reported	HP:0000164
Valvular heart disease (mitral/ aortic thickening, TR)	Clinical sign	Childhood	Progressive	Reported	HP:0001654
Ventricular septal defect	Structural	Congenital	—	1 patient	HP:0001629

Phenotype	Type	Onset	Severity / Course	Frequency	Suggested HPO
Elevated plasma lysosomal enzymes	Lab abnormality	Congenital	—	Where tested	HP:0003128
Abnormal urinary GAGs	Lab abnormality	Congenital	—	Where tested	Mucopolysacchariduria
Premature death	Outcome	2nd–3rd decade (or childhood)	Fatal	Reported	HP:0001522

Quality-of-life impact. Severe. Progressive loss of mobility, marked short stature, skeletal deformity, cardiorespiratory involvement, and premature death impose profound functional impairment and dependency. No disease-specific QoL instrument (EQ-5D/SF-36/PROMIS) data exist for DMAN given its rarity.

Section 4 — Genetic / Molecular Information

Causal gene. *LYSET* / *TMEM251* (alias *GCAF*, *C14orf109*, *UPF0694*); HGNC:20218; gene OMIM 619332; locus 14q32.12 (*Ensembl* ENSG00000153485; *GRCh38* chr14:93,184,951–93,188,463, plus strand). A processed pseudogene, *TMEM251P*, lies on chr5 within an *ADGRV1** intron.

Gene product. UniProt Q8N6I4: a **169-amino-acid** protein (long isoform, ~18.7 kDa, pI ≈ 8.4) containing a **DUF4583** domain (~aa 35–160) and **two transmembrane helices**, with recent experimental consensus that both N- and C-termini face the **cytosol** (PMID: 39677738). Two splice isoforms exist.

Pathogenic variant spectrum. All reported disease alleles are **biallelic loss-of-function** (nonsense, frameshift, or a conserved-residue missense). Variants are classified **pathogenic/likely pathogenic** under ACMG/AMP criteria (null variants in a LoF-intolerant, conserved gene; segregation in consanguineous families; functional concordance with M6P-pathway loss). All are **germline**; no somatic disease association exists for this Mendelian condition.

Variant (cDNA)	Protein	Type	Family / origin	Reference
c.133C>T	p. (Arg45Trp)	Missense	Family NMD02 (consanguineous)	PMID: 33252156
c.215dupA	p. (Tyr72Ter)	Frameshift/ nonsense	Family ID01 (consanguineous)	PMID: 33252156
c.197dupA	frameshift	Frameshift	Two Iranian brothers	PMID: 40171858
c.112C>T (NM_001098621.4)	p. (Gln38Ter)	Nonsense	Brazilian (first Western) patient	PMID: 41858182

Allele frequency. These variants are absent or ultra-rare in population databases (gnomAD), consistent with a severe recessive disorder. No common susceptibility alleles.

Functional consequence. Loss of function — destabilization/mislocalization of GNPT and global loss of M6P tagging (see Section 6).

Modifier genes. None formally established for DMAN. Mechanistically plausible modifiers within the same pathway include **GNPTAB/GNPTG** (the phosphotransferase subunits), **GOLPH3/GOLPH3L** (retain the LYSET–GNPT complex in the *cis*-Golgi; [PMID: 39587297](#)), **MBTPS1/Site-1-Protease** (GNPT cleavage), and **M6PR/IGF2R** (receptor delivery).

Epigenetic information / chromosomal abnormalities. None reported. DMAN is a single-gene point/indel disorder; no methylation signature, large structural variant, aneuploidy, translocation, or CNV mechanism has been described.

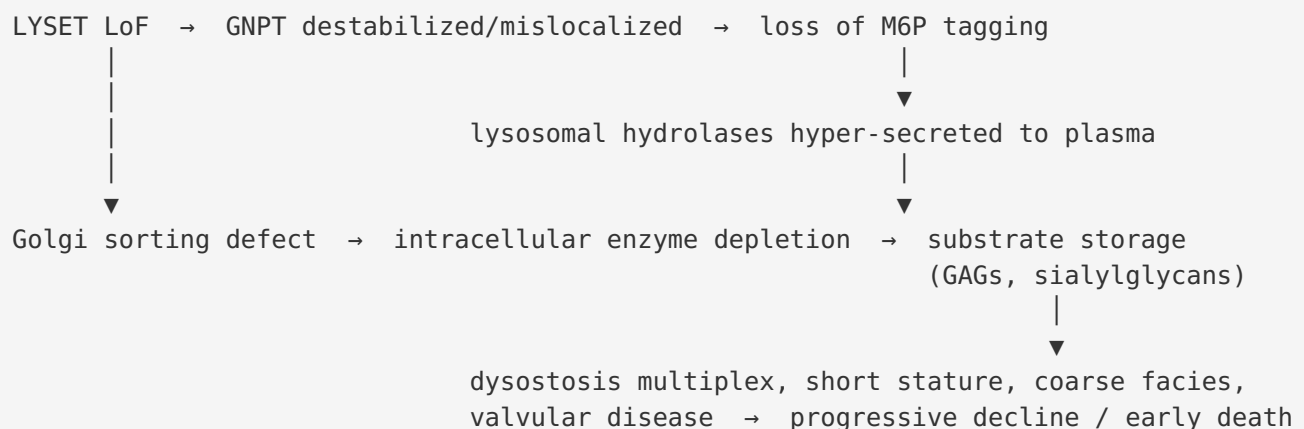
Section 5 — Environmental Information

Not applicable. DMAN is a purely genetic Mendelian disease. There are **no known environmental factors, lifestyle factors, or infectious agents** that cause or trigger it. (Of note, LYSET is separately required for entry of cathepsin-dependent viruses including SARS-CoV-2 — see Section 6 — but this is a property of the pathway, not an environmental cause of DMAN.)

Section 6 — Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic loss-of-function variants in *LYSET/TMEM251*** result in absent or nonfunctional LYSET protein in the Golgi. (*demonstrated* — [PMID: 33252156](#))
2. Loss of LYSET **destabilizes GlcNAc-1-phosphotransferase (GNPT)**, whose hydrophilic transmembrane domain requires LYSET for retention; GNPT is **mislocalized from the cis-Golgi to the lysosome** and fails to be properly cleaved by Site-1-Protease. (*demonstrated* — [PMID: 36074822](#), [PMID: 36074821](#), [PMID: 39677738](#))
3. Loss of GNPT activity leads to **global loss of mannose-6-phosphate (M6P) tagging** of the ~60 soluble lysosomal hydrolases. (*demonstrated* — [PMID: 36074821](#))
4. Untagged hydrolases are **not captured by M6P receptors**, so instead of being routed to the lysosome they are **hyper-secreted into the extracellular space / plasma**. (*demonstrated* — *elevated plasma enzymes*: [PMID: 41858182](#))
5. Cells become **depleted of intracellular lysosomal enzymes** → impaired turnover of macropinocytic and autophagic cargoes → **accumulation of undigested substrate (glycosaminoglycans, sialylated glycoconjugates)** = a **lysosomal storage state ("I-cell"/inclusion-cell phenotype)**. (*demonstrated* — [PMID: 36074822](#))
6. Storage in **chondrocytes, osteoblasts, fibroblasts, and cardiac/valvular tissue** disrupts bone/cartilage growth and connective-tissue homeostasis. (*inferred from *MLII* pathology + patient biochemistry*)
7. This produces the **clinical manifestations**: dysostosis multiplex, extreme short stature, coarse facies, valvular/cardiac disease, and progressive multisystem decline leading to premature death. (*demonstrated clinically* — [PMID: 33252156](#))



Detail

Molecular pathway. The **mannose-6-phosphate (M6P) lysosomal enzyme-trafficking pathway** (KEGG/Reactome "lysosome"; N-glycan/M6P biosynthesis). LYSET was independently identified by genome-scale CRISPR and nutrient-selection screens as the key **regulator of the M6P pathway**, acting via GNPT ([PMID: 36074821](#), [PMID: 36074822](#)):

"Without LYSET, GlcNAc-1-phosphotransferase was unstable because of a hydrophilic transmembrane domain. Consequently, LYSET-deficient cells were depleted of lysosomal enzymes and impaired in turnover of macropinocytic and autophagic cargoes." ([PMID: 36074822](#))

"LYSET deficiency resulted in global loss of M6P tagging and mislocalization of GlcNAc-1-phosphotransferase from the Golgi complex to lysosomes." ([PMID: 36074821](#))

"TMEM251 is a two-transmembrane protein indispensable for GNPT stability, cleavage by Site-1-Protease (S1P), and enzymatic activity." ([PMID: 39677738](#))

Cellular processes (GO). Protein targeting to lysosome (GO:0006622 / GO:0007041), Golgi organization (GO:0007030), autophagy (GO:0006914), and defective substrate catabolism. LYSET interacts with **GOLPH3/GOLPH3L** and retromer to retain the LYSET–GNPT complex in the *cis*-Golgi ([PMID: 39587297](#)).

Protein dysfunction. Loss of function at two levels: (i) loss of LYSET stabilizer function; (ii) consequent destabilization and mislocalization of GNPT. Upstream lesion = LYSET; downstream effectors = GNPT → all M6P-dependent hydrolases.

Biochemical abnormalities. The pathognomonic "I-cell" pattern was directly demonstrated in a DMAN/LYSET patient ([PMID: 41858182](#)):

"Nine enzymes showed increased levels in plasma, and seven showed decreased levels in fibroblasts. Abnormal sialic acid profile and GAGs (glycosaminoglycans) were detected in urine."

Convergence with mucopolipidosis II. DMAN and MLII share the identical downstream defect ([PMID: 36074821](#), [PMID: 36074822](#)):

"GlcNAc-1-phosphotransferase deficiency leads to the severe lysosomal storage disorder mucopolidosis II (MLII)." (PMID: 36074821)

"LYSET represents a core component of the lysosomal enzyme trafficking pathway, underlies the pathomechanism for hereditary lysosomal storage disorders, and may represent a target to suppress metabolic adaptations in cancer." (PMID: 36074822)

MLII patients share the clinical hallmarks (PMID: 33000604):

"Patients with MLII alpha/beta present coarse facial features, cessation of statural growth, important skeletal manifestations, impaired neuromotor development and cardiorespiratory involvement."

Pleiotropy of LYSET. Beyond DMAN, LYSET is a hub with two additional, mechanistically-linked roles (PMID: 36074821, PMID: 36074822):

"We used genome-scale CRISPR screens to identify lysosomal enzyme trafficking factor (LYSET, also named TMEM251) as essential for infection by cathepsin-dependent viruses including severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2)." (PMID: 36074821)

"Mammalian cells can generate amino acids through macropinocytosis and lysosomal breakdown of extracellular proteins, which is exploited by cancer cells to grow in nutrient-poor tumors." (PMID: 36074822)

Suggested ontology terms. GO biological process: GO:0006622 (protein targeting to lysosome), GO:0016192 (vesicle-mediated transport), GO:0006914 (autophagy). GO cellular component: GO:0005794 (Golgi apparatus), GO:0000139 (Golgi membrane), GO:0005764 (lysosome). CHEBI: mannose-6-phosphate, glycosaminoglycan. CL cell types: chondrocyte (CL:0000138), osteoblast (CL:0000062), fibroblast (CL:0000057).

Section 7 — Anatomical Structures Affected

Organ level. - Primary: Skeleton (bone and cartilage) — dysostosis multiplex, short stature, kyphoscoliosis, hip dysplasia, craniosynostosis (UBERON:0001474 bone; UBERON:0002418 cartilage). - **Secondary:** Heart / cardiac valves (valvular thickening, tricuspid regurgitation, VSD, dilated-cardiomyopathy risk as in MLII; UBERON:0000948 heart, UBERON:0000946 cardiac valve); respiratory tract/airway (cause of death; airway distortion as in MLII); abdominal organs (protruding abdomen, possible organomegaly); teeth (UBERON:0001091); connective tissue broadly. - **Body systems:** Musculoskeletal (primary), cardiovascular, respiratory.

Tissue / cell level. Connective tissue and cartilage are principally affected. Key storage-affected cell populations: **chondrocytes (CL:0000138)**, **osteoblasts (CL:0000062)**, and **fibroblasts (CL:0000057)** — the last directly shown to have reduced intracellular hydrolases and storage inclusions in patients.

Subcellular level. The **Golgi apparatus (GO:0005794 / GO:0000139)** is the site of the primary lesion; the **lysosome (GO:0005764)** is the site of pathological substrate accumulation. Autophagosomal cargo turnover is impaired.

Localization / lateralization. Skeletal involvement is **generalized and bilateral/symmetric** (a systemic storage disorder), as expected for a metabolic dysplasia rather than a focal malformation.

Section 8 — Temporal Development

Onset. Congenital/early-infantile. Coarse facies documented from ~5 months of age; short stature and dysostosis evident in infancy/early childhood ([PMID: 41858182](#), [PMID: 33252156](#)). Onset pattern is **insidious and chronic-progressive**.

Progression. Progressive and lifelong. Skeletal changes worsen over time, patients "gradually lost mobility," and facial coarsening increases with age ([PMID: 33252156](#)). No remission or relapsing-remitting pattern; the course is a monotonic decline.

Disease duration / critical windows. Chronic and ultimately fatal. Reported deaths: two oldest patients in their **twenties**; one child at **age 5**; the Brazilian patient died from respiratory causes. Any therapeutic window is presumed to be **early** (pre-symptomatic/early-symptomatic), by analogy to MLII, before irreversible skeletal and airway remodeling.

Section 9 — Inheritance and Population

Inheritance pattern. Autosomal recessive (biallelic LoF), OMIM #619345.

Penetrance / expressivity. Penetrance appears **complete** in reported biallelic individuals; expressivity ranges from severe childhood death to survival into the twenties, indicating **variable expressivity** (partly variant-dependent — e.g., residual function of missense p.Arg45Trp vs. null alleles).

Epidemiology. Ultra-rare. Only ~4 families / 6+ patients are described worldwide; no prevalence or incidence figures are available. Prevalence is effectively unquantified (well below any registry threshold).

Consanguinity / founder effects. Consanguinity is prominent — the founding families and the Iranian sibling pair were consanguineous, consistent with rare recessive homozygosity. No broad founder allele is established; variants appear private to families.

Carrier frequency. Not established; expected to be extremely low given absence/ultra-rarity in gnomAD.

Population demographics. Reported patients span **Middle Eastern (consanguineous), Iranian, and Brazilian (first Western)** backgrounds ([PMID: 40171858](#), [PMID: 41858182](#)). **Sex ratio** appears balanced (both sexes affected); no age or geographic clustering beyond consanguineous populations.

Section 10 — Diagnostics

Biochemical/laboratory testing (the diagnostic key). The disorder produces a **mucopolidosis-II-like biochemical signature**: multiple lysosomal hydrolases **elevated in plasma/serum** with **reduced activity in cultured fibroblasts/leukocytes**, plus **abnormal urinary glycosaminoglycans and sialic acid** ([PMID: 41858182](#)):

"Nine enzymes showed increased levels in plasma, and seven showed decreased levels in fibroblasts. Abnormal sialic acid profile and GAGs (glycosaminoglycans) were detected in urine."

Imaging. Skeletal survey (X-ray) shows **dysostosis multiplex** (radiographic hallmark). CT/MRI as needed for craniosynostosis, spine (kyphoscoliosis, hemivertebrae), and hips; echocardiography for valvular disease.

Genetic testing (definitive). Molecular confirmation of biallelic *LYSET*/*TMEM251* variants.

Because DMAN mimics MLII/III, *LYSET* should be included on MLII/III-like differential panels (PMID: 40171858):

"We propose the term 'LYSET-related mucopolipidosis' to describe this disorder and emphasize the importance of including *LYSET* in the genetic diagnostic panel for MLII/III-like presentations."

Recommended approach: **WES or WGS** (unbiased, first-line for undiagnosed skeletal dysplasia); **targeted panels** for mucopolipidosis/dysostosis that include *GNPTAB*, *GNPTG*, and *LYSET*; **single-gene *LYSET*** testing when biochemistry is I-cell-like and a family variant is known. Chromosomal microarray/karyotype are **not informative** (no CNV/structural mechanism).

Clinical criteria / differential diagnosis. No formal consensus criteria exist. Diagnosis rests on the triad of (1) dysostosis multiplex + extreme short stature + coarse facies, (2) I-cell biochemistry, (3) biallelic *LYSET* variants. Key **differentials**: mucopolipidosis II/III (*GNPTAB*/*GNPTG*), mucopolysaccharidoses (MPS I–VII; distinguished by specific enzyme deficiencies and urinary GAG patterns), oligosaccharidoses (α/β -mannosidosis, fucosidosis, sialidosis, galactosialidosis), and multiple sulfatase deficiency. *LYSET* testing distinguishes DMAN from GNPT-intrinsic MLII.

Screening. No newborn or population screening exists. **Cascade/carrier testing** of at-risk relatives in affected consanguineous families is appropriate once the familial variant is known.

Section 11 — Outcome / Prognosis

Prognosis is poor. DMAN is **progressive and life-limiting**. Reported mortality: two oldest patients died in their **twenties**, one child at **age 5**, and the first Western patient died of **respiratory causes** (PMID: 33252156, PMID: 41858182). No formal 5-/10-year survival statistics exist given the tiny cohort.

Morbidity/function. High disability: progressive **loss of ambulation**, severe short stature, skeletal deformity, and cardiorespiratory compromise. By analogy to MLII, **airway distortion (saber-sheath trachea, fixed narrowing)** and **cardiac/valvular disease (including dilated cardiomyopathy)** are major morbidity/mortality drivers ([PMID: 32270604](#), [PMID: 33000604](#)).

Complications. Recurrent respiratory infections/aspiration, airway obstruction, valvular heart disease/heart failure, and progressive orthopedic deformity.

Prognostic factors. Likely determined by **residual GNPT activity** (variant severity), extent of cardiorespiratory involvement, and age at diagnosis. No validated prognostic biomarkers exist, though plasma hydrolase levels and urinary GAG/sialic acid could serve as candidate biochemical monitors.

Section 12 — Treatment

There is no disease-specific or curative therapy for DMAN. Management is **supportive/symptomatic**, and disease-modifying candidates are **extrapolated from mucopolidosis II** (mechanistically identical GNPT–M6P defect). NCIT-suggested intervention terms are noted below.

Supportive / symptomatic care (standard of care)

Multidisciplinary management modeled on MLII: respiratory support (**CPAP**, antibiotics for infections/aspiration), cardiac management for valvular disease (**ACE inhibitor — captopril; diuretic — spironolactone**), orthopedic/spine care, nutrition, physical/occupational/speech therapy, and genetic counseling ([PMID: 41830382](#)). (*NCIT: Supportive Care; ACE Inhibitor; Diuretic Therapy*)

Candidate disease-modifying strategies (from MLII research)

Strategy	Evidence (in MLII / model)	Key result	Reference
Hematopoietic stem cell transplantation (HSCT)	MLII patients	Sustained reduction of lysosomal storage and bone-metabolism markers; some motor/cognitive gains — but skeletal disease progressed, airway distortion persisted, benefit unclear	PMID: 33505859 , PMID: 37484777 , PMID: 32270604
Zoledronic acid (bisphosphonate)	<i>Gnptab</i> p.R364X knock-in MLII mouse	Increased M6P-dependent and -independent luminal hydrolases; reduced GAG storage; improved bone density, cartilage, motor function, hepatomegaly, brain neuroinflammation	PMID: 42320387
Enzyme cross-correction (M6PR/IGF2R double-KO supernatant supplying M6P-tagged enzymes)	MLII fibroblasts/ patient cells	Restored lysosomal enzyme activity, reduced inclusion bodies, improved autophagy — proof of concept	PMID: 42665096
ASO exon-skipping (GNPTAB)	MLII fibroblasts / HEK293T	Modest, variant-specific benefit; did not fully restore GlcNAc-PT activity	PMID: 42310785

Direct evidence quotes.

"HSCT resulted in a sustained reduction of lysosomal storage and bone metabolism markers." ([PMID: 33505859](#))

"Zoledronic acid treatment increased both M6P-dependent and M6P-independent luminal hydrolases, and reduced glycosaminoglycan storage in MLII skin fibroblasts." ([PMID: 42320387](#))

Important caveat. DMAN is a **trafficking (M6P-tagging) disorder, not a single-enzyme deficiency**, so conventional single-enzyme **enzyme replacement therapy is not applicable**; therapies must restore the tagging/trafficking machinery or supply pre-tagged enzymes (cross-correction). No genotype-guided pharmacogenomic protocol exists. (*NCIT: Bone Marrow/Stem Cell Transplantation; Bisphosphonate Therapy — Zoledronic Acid; Antisense Oligonucleotide Therapy; Gene Therapy [investigational].*)

Section 13 — Prevention

- **Primary prevention:** Not applicable to disease occurrence in a conceived affected individual. At the population/family level, **genetic counseling** for consanguineous couples and **carrier/cascade testing** once a familial *LYSET* variant is identified are the principal preventive measures.
 - **Reproductive options:** For couples with a known biallelic risk, **prenatal diagnosis** and **preimplantation genetic testing (PGT-M)** are options.
 - **Secondary prevention:** Early diagnosis (biochemical + molecular) enables anticipatory management of cardiorespiratory and skeletal complications.
 - **Tertiary prevention:** Proactive respiratory care (CPAP, infection prophylaxis), cardiac surveillance/echocardiography, and orthopedic monitoring to limit complications.
 - **Immunization / public health / environmental interventions:** Not applicable (no infectious or environmental etiology). Routine childhood immunization and aggressive treatment of respiratory infections are prudent given airway vulnerability.
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Section 14 — Other Species / Natural Disease

- **Taxonomy / orthologs.** *LYSET/TMEM251* is **evolutionarily conserved** (PMID: 33252156), with orthologs in mouse (*Tmem251*), zebrafish (*tmem251*), and other vertebrates. NCBI Taxon IDs of relevance: *Homo sapiens* (9606), *Mus musculus* (10090), *Danio rerio* (7955).
- **Natural disease in animals.** No **naturally occurring** *LYSET*-related disease has been reported in companion animals or wildlife (no OMIA entry noted). The related GNPT/M6P disorder mucopolidosis has veterinary analogues, but a spontaneous *LYSET* disorder is not documented.

- **Comparative biology.** The M6P pathway and the LYSET–GNPT interaction are conserved across vertebrates, so disease mechanisms are expected to translate; the zebrafish knockout recapitulates skeletal and cardiac pathology (Section 15).
- **Transmission / zoonosis.** Not applicable (genetic, non-transmissible).

Section 15 — Model Organisms

Zebrafish (*Danio rerio*) *tmem251/lyset* knockout — principal validated model. In a genome-wide CRISPR screen, TMEM251 was identified as the first regulator of M6P modification; its deletion causes mistargeting of most lysosomal enzymes and accumulation of undigested material. Crucially, the whole-animal knockout **phenocopies mucopolidosis II** ([PMID: 36096887](#)):

"In zebrafish, TMEM251 deletion leads to severe developmental defects including heart edema and skeletal dysplasia, which phenocopies Mucopolidosis Type II."

"We name TMEM251 as GNPTAB cleavage and activity factor (GCAF) and its related disease as Mucopolidosis Type V."

This model recapitulates the two most clinically important axes of DMAN — **skeletal dysplasia and cardiac disease** — making it valuable for mechanistic and preclinical therapeutic work. **Limitation:** developmental fish models may not capture the slowly progressive skeletal/airway remodeling and adult mortality seen in humans.

Cellular / in vitro models. Extensive: genome-scale CRISPR-knockout human cell lines that revealed the mechanism ([PMID: 36074821](#), [PMID: 36074822](#), [PMID: 36096887](#), [PMID: 39587297](#)); **patient-derived fibroblasts** demonstrating the I-cell biochemistry ([PMID: 41858182](#)). These support functional-genomics, trafficking, and cross-correction studies.

Mouse. No dedicated *Lyset/Tmem251* DMAN mouse is reported here; however, the mechanistically identical **Gnptab p.R364X knock-in MLII mouse** is a directly relevant surrogate for testing pathway-restoring therapy (zoledronic acid) ([PMID: 42320387](#)). A *Lyset* knockout mouse is an obvious high-value gap (see Follow-up).

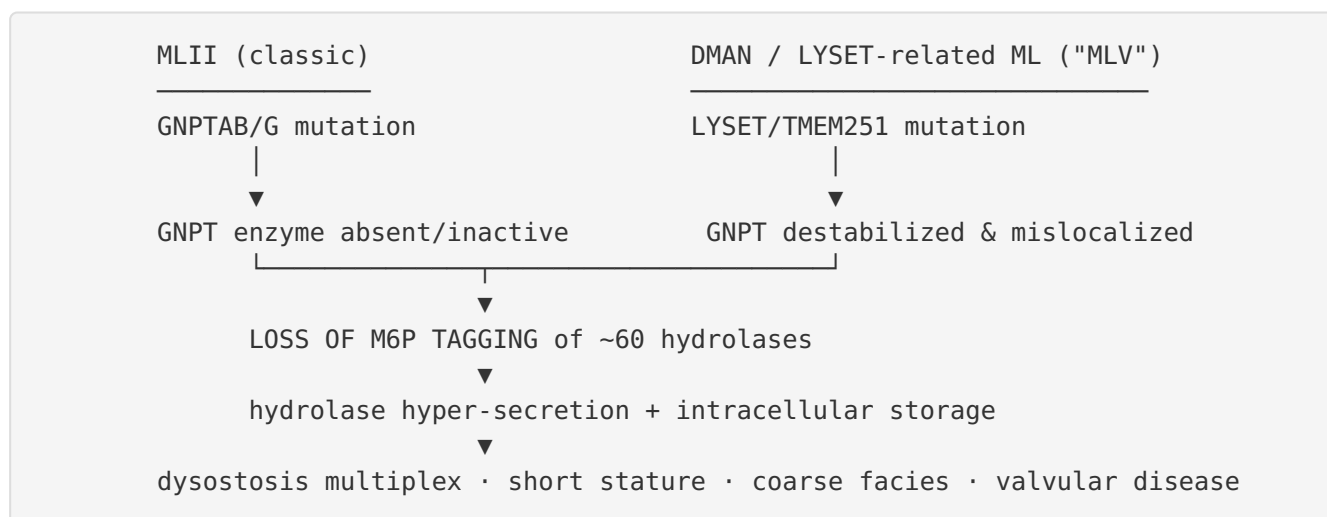
Model databases: ZFIN (zebrafish), MGI/IMPC (mouse), Cellosaurus (patient fibroblast lines).

Mechanistic Model / Interpretation

DMAN is best understood as a **"Golgipathy"** — a disease of the Golgi sorting machinery ([PMID: 36456556](#)):

"Association genetic studies and genome-scale CRISPR screens have recently identified ARF3 and TMEM251/LYSET/GCAF as Golgi-resident factors essential to brain and skeletal development."

The disease sits one step **upstream in the same pathway** as classic mucopolidosis II. Whereas MLII results from intrinsic loss of GNPT (the enzyme that writes the M6P tag), DMAN results from loss of **LYSET, the factor that keeps GNPT stable and correctly localized**. Both converge on the same endpoint — no M6P tag, no lysosomal delivery, hydrolase hyper-secretion, and intracellular storage — which is why DMAN is biochemically and clinically an "I-cell disease."



This convergence is the **central actionable insight**: the substantial MLII research base (natural history, biomarkers, HSCT, bisphosphonates, cross-correction, ASOs) is directly transferable to DMAN, and DMAN patients should be managed and studied within the mucopolidosis framework.

Evidence Base

PMID	Contribution	Evidence type
33252156	Founding report: biallelic <i>TMEM251</i> variants (p.Arg45Trp, p.Tyr72Ter); core phenotype & mortality	Human clinical / genetics
36074821	LYSET essential for M6P tagging; GNPT mislocalization; viral-entry role	In vitro / CRISPR
36074822	GNPT instability without LYSET; enzyme depletion; cancer nutrient scavenging	In vitro / CRISPR
39677738	TMEM251 required for GNPT stability, S1P cleavage, activity; topology	In vitro / structural
36096887	Zebrafish knockout phenocopies MLII; "GCAF"/"Mucopolipidosis Type V"	Model organism
40171858	Two Iranian brothers (c.197dupA); "LYSET-related mucopolipidosis"; panel inclusion	Human clinical
41858182	First Western patient (p.Gln38Ter); direct I-cell biochemistry	Human clinical / biochemistry
33000604	MLII clinical overlap (coarse facies, growth arrest, cardiorespiratory)	Human clinical (MLII)
36456556	Golgipathy classification	Review
39587297	GOLPH3/GOLPH3L retain LYSET–GNPT in Golgi	In vitro
33505859	HSCT partial benefit in MLII	Human clinical (MLII)
42320387	Zoledronic acid restores hydrolases in MLII mouse	Model organism (MLII)
42665096	M6PR/IGF2R dKO cross-correction of MLII cells	In vitro (MLII)
42310785	GNPTAB ASO exon-skipping (modest)	In vitro (MLII)
41830382	MLII symptomatic care (CPAP, captopril, spironolactone)	Human clinical (MLII)
32270604	Airway distortion persists post-HSCT in MLII	Human clinical (MLII)

PMID	Contribution	Evidence type
37484777	HSCT outcomes in MLII (unclear benefit)	Human clinical (MLII)

Limitations and Knowledge Gaps

1. **Extremely small cohort.** Only ~4 families / 6+ patients are described; there are **no prevalence, incidence, survival, or QoL statistics**, and natural history is anecdotal.
2. **Genotype–phenotype correlation is undefined.** Whether the missense p.Arg45Trp retains residual function (milder course) versus null alleles is unproven.
3. **Therapeutics are entirely extrapolated from MLII.** No DMAN-specific trial or treatment outcome data exist; HSCT benefit is unclear even in MLII, and skeletal/airway disease progresses.
4. **No mammalian (mouse) DMAN model.** Preclinical therapy testing currently relies on zebrafish and on MLII (*Gnptab*) mouse surrogates.
5. **CNS/neurodevelopmental extent under-characterized.** MLII and related glycoproteinoses feature neurodevelopmental involvement; the neurodevelopmental burden in DMAN is not well quantified.
6. **No validated biomarkers** for monitoring disease progression or treatment response specific to DMAN.

Proposed Follow-up Experiments / Actions

1. **Generate a *Lyset/Tmem251* knock-in or conditional-knockout mouse** carrying a patient variant (e.g., p.Arg45Trp) to model progressive skeletal/cardiac disease and serve as a preclinical therapeutic platform.
2. **Test pathway-restoring therapies directly in LYSET-deficient systems:** zoledronic acid, and M6PR/IGF2R-dKO cross-correction supernatant, in patient-derived fibroblasts and the zebrafish knockout, to confirm transferability from MLII.

3. **Establish an international DMAN/LYSET registry** with standardized biochemical (plasma/fibroblast hydrolase panels, urinary GAG/sialic acid), radiographic, and cardiorespiratory phenotyping to build natural history and candidate biomarkers.
4. **Add *LYSET* to all mucopolidosis/dysostosis diagnostic gene panels** and reflex-test I-cell-biochemistry patients who are *GNPTAB*/*GNPTG*-negative, to find undiagnosed cases.
5. **Systematic genotype–function mapping** of *LYSET* variants (missense vs. null) using GNPT-stability and M6P-tagging readouts to predict severity and stratify future trials.
6. **Cardiorespiratory surveillance protocol** (serial echocardiography, airway imaging) developed for DMAN based on MLII experience, given these are the leading causes of death.

Evidence source key: Human clinical/genetic (patient reports); Model organism (zebrafish; MLII mouse); In vitro (CRISPR screens, patient fibroblasts, cell lines); Review/computational. Where DMAN-specific data were unavailable, mechanistically identical mucopolidosis II evidence is clearly labeled as such.
